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Q1) Attempt the following.

a) what is Error in network data transmission?
Explain types of errors with example.

Error in network data transmission:

During the transmission data may be corrupted due to some reason such as noise, cross talk etc.

For example:

Sender sends: 10110011

Receiver gets: 10100011

Here one bit changed → error occurred.

Types of Errors:

There may be 3 types of error.

Single bit error:

Frame is received with only one corrupt bit.

Sent → Received.

10110011

10110111

Multiple bits error:

Frame is received with more than one bits in corrupted state.

Sent → Received.

10110011

10100111

Burst error:

Frame is received, which contains more than 2 consecutive bits are corrupted.

sent	→	Received
10110011		1 <u>100</u> 0111

b) Determine whether the received data is accepted or not using the checksum method.
0101010101011010101011011011 and the checksum is 01000101?

original data.

01010101 01011010 10101011 10110011

0	1	0	1	0	1	0	1
0	1	0	1	1	0	1	0
1	0	1	0	1	0	1	1
1	0	1	1	0	0	1	1

1 ⇒ 01010101

2 ⇒ 01011010

3 ⇒ 10101011

4 ⇒ 10110011

① + ②

0	1	0	1	0	1	0	1
0	1	0	1	1	0	1	0
1	0	1	0	1	1	1	1

(no carry)

③ ⇒ 10101011

1	0	1	0	1	0	1	1
1	0	0	0	0	0	0	1
0	1	0	1	1	0	1	1

④ ⇒ 10110011

1	0	1	1	0	0	1	1
1	0	0	0	0	1	1	0
0	0	0	0	0	0	0	1

$$\textcircled{1} + \textcircled{2} + \textcircled{3} + \textcircled{4} \Rightarrow 00001111$$

Add check sum:

$$\textcircled{1} + \textcircled{2} + \textcircled{3} + \textcircled{4} \Rightarrow 00001111$$

$$\text{check sum} \Rightarrow \begin{array}{r} 01000101 \\ 01010100 \\ \hline \end{array} //$$

$$\Rightarrow 01010100 \neq 11111111$$

It's not all 1s.

Final answer is. Received. data was Rejected.

c). If the data unit to be transmitted is.

10101001 00111001 the find the checksum.

and demonstrate how the receiver verifies the received data.

$$\textcircled{1} \Rightarrow \begin{array}{r} 11 \\ 10101001 \end{array}$$

$$\textcircled{2} \Rightarrow \begin{array}{r} 00111001 \end{array}$$

$$\textcircled{1} + \textcircled{2} \Rightarrow 11100010 \text{ (no carry)}$$

invert all bit's

$$\textcircled{1} + \textcircled{2} \Rightarrow 11100010 \rightarrow 00011101 //$$

$$\text{check sum} \Rightarrow 00011101$$

Receiver Side verification.

$$\textcircled{1} \Rightarrow 10101001$$

$$\textcircled{2} \Rightarrow \begin{array}{r} 00111001 \end{array}$$

$$\textcircled{1} + \textcircled{2} \Rightarrow 11100010$$

$$\text{check sum} \Rightarrow 00011101$$

$$\begin{array}{r} 11111111 \\ \hline \end{array} //$$

$$\text{check sum} \Rightarrow 00011101$$

Receiver Result: 11111111 Data is Accepted.

Q2) Attempt the following:

- a) Explain the parity and types of parity conventions with examples. Also demonstrate the simple parity check error detection method with example.

Parity:

Parity is an error-detection technique used in data transmission where an extra bit (called the parity bit) is added to data to make the total number of 1s either even or odd.

It helps the receiver detect whether the transmitted data has been altered due to noise or errors.

Types of parity:

Even parity:

The parity bit is chosen such that the total number of 1s become even.
example:

Data: 1010001

Number of 1s: 3 (odd)

To make it ~~transmitted~~ even parity bit: 1

Transmitted Data: 1010001 ** 1 ** (Even) parity

odd parity:

The parity bit is chosen such that the total number of 1s become odd.

example:

Data: 1100101

Number of 1s: 4 (even)

To make it odd parity bit: 1

Transmitted Data: 1100101 ** 1 ** (odd) parity

Simple parity check:

Simple parity check allows the receiver to detect single-bit errors during transmission.

- i) Count the number of 1s in the data.
- ii) Adds a parity bit (even or odd)

Receiver:

- i) Count the total number of 1s (including parity bit).
- ii). If parity rule is satisfied $\rightarrow$ Accepted.
- iii). If parity rule is broken $\rightarrow$ Error detected.

Example of Error detection using (Even parity).

Sender:

Data: 1010110

Number of 1s: 4 (Even)

Even parity $\rightarrow$ parity bit = 0

Transmitted = 10101100

Receiver:

Assume a single bit error occurs during transmission.

Received Data = 10111100

Count the no. of 1(s)

10111100 = has (5) ones (odd)

Expected (even parity) but receiver got (odd)
Error Detected //

Example of error detection using (Odd parity).

Sender:

Data: 1001101

Number of 1s: 4 (Even)

odd parity $\rightarrow$ parity bit = 1

Transmitted = 10011011

Received Total = 10011011

Total 1(s) = 5 (odd) $\rightarrow$ matches odd parity

data is Accepted (no error) //

- parity adds 1 bit to detect errors.
- Even parity $\rightarrow$ total 1s must be even.
- odd parity $\rightarrow$ total 1s must be odd.
- simple parity checks detect single bit errors only.

b) Using the CRC technique of error detection, find the actual transmitted data where the data word to be sent is 11001001 and generator key is 1101.

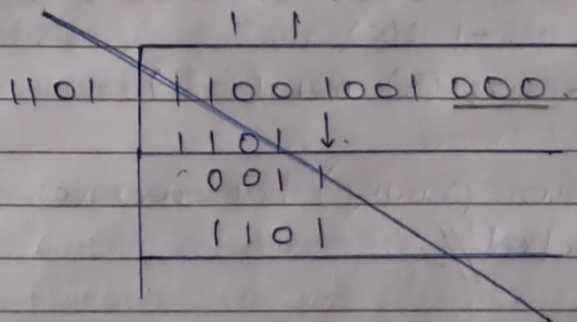
CRC in sender side:

Data word: 11001001

Generator (divisor) : 1101 (length = 4)

$$\therefore \text{degree} = 4 - 1 = 3 //$$

Append 3 zero to the word.



$$\begin{array}{r}
 1101 \overline{) 11001001000} \\
 \underline{1101} \\
 00011001000 \\
 \underline{1101} \\
 00000011000 \\
 \underline{1101} \\
 00000000010 \rightarrow \text{Remainder}
 \end{array}$$

When the leftmost bit of the remainder is zero, we use 0000 instead of original divisor.

CRC in Receiver side:

CRC Received: 010.

Data word: 11001001.

Generator (divisor): 1101.

$$\begin{array}{r}
 1101 \overline{) 11001001010} \\
 \underline{1101} \\
 00011001010 \\
 \underline{1101} \\
 00000011010 \\
 \underline{1101} \\
 00000000000 \rightarrow \text{Remainder}
 \end{array}$$

Data plus CRC Received.

Find the remainder after dividing received frame. = 000
 All zero mean no error detected - receiver accept the frame.

- c) if the code word received at the receiver side is 110100001 and the generator polynomial is $x^3 + x^2 + 1$. then using the CRC technique of error detection check whether the data is to be accepted or not.

code word received at the receiver side.
110100001 which mean CRC Bit already added.

poly nomial is $x^3 + x^2 + 1$.

$$x^3 \rightarrow 1$$

$$x^2 \rightarrow 1$$

$$x \rightarrow 0$$

$$x^0 \rightarrow 1 \text{ (constant)}.$$

generator (divisor) = 1101 //

$$\begin{array}{r|l} 1101 & 110100001 \\ & \underline{1101} \\ & 000000001 \rightarrow \end{array}$$

Not all are zero mean error detected.
data was not accepted and reject the received data.

Q3) Attempt the following.

- a) What is error correction? demonstrate how hamming code technique is used to detect and correct the error.

Error Correction:

Error correction is a technique used data transmission and communication system to detect and correct errors that occurs during transmission.

When data travels over a noisy channel. (eg. wireless network, copper cable) some bits may change from $0 \rightarrow 1$ or $1 \rightarrow 0$.

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Error correcting codes add extra redundant bits (parity bits) to the original data so that the receiver can identify and even fix the wrong bit.

Types of Correction.

Forward error correction:

The sender adds enough redundant bits so the receiver can correct errors without retransmission.

Backward error correction:

Receiver detects an error and asks the sender to retransmit.

Hamming Code - Detect & correct errors.

Hamming code uses parity bit at positions that are powers of two.

$$1 = 2^0$$

$$2 = 2^1$$

$$4 = 2^2$$

X $8 = 2^3$ (not needed for 7-bit code).

position	Binary
1	001
2	010
3	011
4	100
5	101
6	110
7	111

$p_1 \Rightarrow$ Least significant bit (1) = 1, 3, 5, 7.

$p_2 \Rightarrow$ 2nd bit (1) = 2, 3, 6, 7.

$p_4 \Rightarrow$ 3rd bit (1) = 4, 5, 6, 7.

Ques Mean ,

Total bit = 7.

Data Bit = 4

parity bit = 3

parity bit position = 1, 2, 4.

Data word bit position = 3, 5, 6, 7.

Data bit = 1011

1011 $\Rightarrow$ ~~D3~~ $\rightarrow$ D1 $\rightarrow$ 1 $\rightarrow$ (3)

D2 $\rightarrow$ 0 $\rightarrow$ (5)

D3 $\rightarrow$ 1 $\rightarrow$ (6)

D4 $\rightarrow$ 1 $\rightarrow$ (7)

Assign position:

position	1	2	3	4	5	6	7
type	P1	P2	D1	P4	D2	D3	D4
Bit	?	?	1	?	0	1	1

calculate parity bit:

P1 (covers $\Rightarrow$ 1, 3, 5, 7)

1 $\rightarrow$?

3 $\rightarrow$ 1

5 $\rightarrow$ 0

7 $\rightarrow$ 1

Count 1's (3, 5, 7) $\Rightarrow$ (2) already even.

so P1 = 0//

P2 (covers $\Rightarrow$ 2, 3, 6, 7)

2 $\rightarrow$?

3 $\rightarrow$ 1

6 $\rightarrow$ 1

7 $\rightarrow$ 1

Count 1's (3, 6, 7) $\Rightarrow$ (3) odd.

To make it even $\rightarrow$ P2 $\rightarrow$ 1//

p_4 (Cover $\Rightarrow 4, 5, 6, 7$)

$4 \rightarrow ?$

$5 \rightarrow 0$

$6 \rightarrow 1$

$7 \rightarrow 1$

Count 1's (5, 6, 7) $\Rightarrow$ (2) even

So $p_4 = 0 //$

Final Hamming Code

position	1	2	3	4	5	6	7
type	p_1	p_2	D_1	p_4	D_2	D_3	D_4
bit	0	1	1	0	0	1	1

Encoded hamming code $= 0110011 //$

Assume error occur

Let's say bit 6 flipped from $1 \rightarrow 0$

0110001

$p_1 \Rightarrow (1, 3, 5, 7) \Rightarrow 0, 1, 0, 1 \Rightarrow \text{Even} \Rightarrow 0$

$p_2 \Rightarrow (2, 3, 6, 7) \Rightarrow 1, 1, 0, 1 \Rightarrow \text{odd} \Rightarrow 1$

$p_4 \Rightarrow (4, 5, 6, 7) \Rightarrow 0, 0, 0, 1 \Rightarrow \text{odd} \Rightarrow 1$

$110_2 = 6_{10}$ error in bit position 6

Received bit 6 = 0

corrected bit 6 = 1

0110011 $\Rightarrow$ extract data bit (3, 5, 6, 7): 1011
same as original

b). Encode a 4-bit data, 1011 using hamming code with even parity convention.

Sol:

Hamming code uses parity bit at positions that are powers of two.

$$\left. \begin{array}{l} 2^0 = 1 \\ 2^1 = 2 \\ 2^2 = 4 \\ 2^3 = 8 \end{array} \right\} (1, 2, 4)$$

parity bits are $\Rightarrow p_1, p_2, p_4$

position	binary
1	001
2	010
3	011
4	100
5	101
6	110
7	111

$p1 \Rightarrow$ Least Significance bit $\Rightarrow (1, 3, 5, 7)$

$p_2 \Rightarrow 2^{\text{nd}} \text{ bit}(1) \Rightarrow (2, 3, 6, 7)$

$P_4 \Rightarrow 3^{\text{rd}} \text{ bit}(1) \Rightarrow (4, 5, 6, 7)$

parity bit position $\Rightarrow 1, 2, 4,$

Data bit position $\Rightarrow 3, 5, 6, 7$

Data bit,

$$1011 \Rightarrow D1 \rightarrow 1 \rightarrow (3)$$

$D_2 \rightarrow 0 \rightarrow (5)$

$$D3 \rightarrow 1 \rightarrow (6)$$
$$D_4 \rightarrow 1 \rightarrow (7).$$

Assign Position

position.	1	2	3	4	5	6	7
type	P1	P2	D1	P4	D2	D3	D4
bit	?	?	1	?	0	1	1

calculate even parity:

P1 (covers $\Rightarrow 1, 3, 5, 7$)

1 $\rightarrow$?

3 $\rightarrow$ 1

5 $\rightarrow$ 0

7 $\rightarrow$ 1

count 1's (3, 5, 7) $\Rightarrow$ 2 (already even)

P1 = 0 //

P2 (covers $\Rightarrow 2, 3, 6, 7$)

2 $\rightarrow$?

3 $\rightarrow$ 1

6 $\rightarrow$ 1

7 $\rightarrow$ 1

count 1's (3, 6, 7) $\Rightarrow$ 3 (odd)

To make it P2 even $\rightarrow$ P2 = 1 //

P4 (covers $\Rightarrow 4, 5, 6, 7$)

4 $\rightarrow$?

5 $\rightarrow$ 0

6 $\rightarrow$ 1

7 $\rightarrow$ 1

count 1's (5, 6, 7) $\Rightarrow$ 2 (even)

P4 = 0 //

Final. hamming code.

position.	1	2	3	4	5	6	7
type	P1	P2	D1	P4	D2	D3	D4
bit	0	1	1	0	0	1	1

encoded. hamming code: 0110011 //

c) If the data received at the receiver side 0011001 is sent using hamming code with even parity convention. then verify the data is correct or not?

Sol:

Hamming code uses parity bit at positions that are power of two.

$$\left. \begin{array}{l} 2^0 = 1 \\ 2^1 = 2 \\ 2^2 = 4 \end{array} \right\} (1, 2, 4)$$

$2^3 = 8$ (not needed for 7 bit code)

Parity bit's are $\Rightarrow P_1, P_2, P_4$

position.	Binary
1	001
2	010
3	011
4	100
5	101
6	110
7	111

$P_1 \Rightarrow$ 1st significance Bit $\Rightarrow (1, 3, 5, 7)$

$P_2 \Rightarrow$ 2nd bit (1) $\Rightarrow (2, 3, 6, 7)$

$P_4 \Rightarrow$ 3rd bit (1) $\Rightarrow (4, 5, 6, 7)$

~~Assign position~~

position.	1	2	3	4	5	6	7
type.	P_1	P_2	D_1				
bit.							

Assign. position.

position.	1	2	3	4	5	6	7
bit.	0	0	1	1	0	0	1

P_1 (covers $\Rightarrow 1, 3, 5, 7$)

$1 \rightarrow 0$

$3 \rightarrow 1$

$5 \rightarrow 0$

$7 \rightarrow 1$

Count 1's $\Rightarrow 2$ (even)

$P_1 = 0 //$

P_2 (covers $\Rightarrow 2, 3, 6, 7$)

$2 \rightarrow 0$

$3 \rightarrow 1$

$6 \rightarrow 0$

$7 \rightarrow 1$

Count 1's $\Rightarrow 2$ (even)

$P_2 = 0 //$

P_4 (covers $\Rightarrow 4, 5, 6, 7$)

$4 \rightarrow 1$

$5 \rightarrow 0$

$6 \rightarrow 0$

$7 \rightarrow 1$

Count 1's $\Rightarrow 2$ (even)

$P_4 = 0 //$

The received data 0011001 is correct and has no error.